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Research assignment 1

Data analytics

- ① A database is an organised collection of data that can be stored, accessed and managed easily. Some examples to consider include customer records, student information and bank statements from accounts.
- ② A data warehouse stores large amounts of historical data from various sources. Unlike normal database, it is mainly used for analysis & reporting.
- ③ A smaller part of a data warehouse focused on a specific department/section such as Sales or finance; mainly used by departments.
- ④ A data lake stores large amounts of raw data in different formats. Businesses prefer this when they need flexibility & want to analyse data later.
- ⑤ Structured data :- Organised in rows & columns, example :- Spreadsheets

Semi structured data :- has some organisation but not fixed tables.

example :- JSON

Unstructured data :- has no fixed structure
Example :- Video, images

⑥ Data modelling is planning how data will be organised and related. it helps create efficient Systems

- * Two types of data modeling
 - Conceptual
 - logical

⑦ Data mining is finding useful patterns & relationships in large datasets. Examples include

- Detecting credit card fraud
- Identifying products customers often buy together

⑧ A DBMS is a Software ~~are~~ used to create, store and manage databases. Examples :-

- * MySQL :- Websites & App's
- * PostgreSQL :- Complex App's
- * Oracle :- Large businesses
- * MongoDB :- Applications using flexible document data

⑨ ETL means Extract; Transform; Load. Data is collected from sources, ~~are~~ cleaned/changed into a useful format, and then loaded into a database or Warehouse. It prepares data for analysis.

⑩ Common Data Storage formats

- * CSV :- Simple table format, good for Spreadsheets
- * JSON :- Stores structured data, commonly used with API's
- * Parquet :- efficient for large datasets & analysis
- * Avro :- Useful for storing & transferring large amounts of structured data.

Section B

⑪ What is Artificial Intelligence?

A technology that allows computers to perform tasks that normally require human intelligence.

- Narrow AI :- Performs tasks like voice assistants
- General AI :- Perform many intellectual tasks like a human & does not currently exist.

⑫ Machine learning allows computers to learn patterns from data instead of being given every rule.

- * Supervised :- learns from labelled data
- * Unsupervised :- finds patterns in unlabelled data
- * Reinforcement :- learns through rewards & penalties

⑬ Deep learning is a type of machine learning that uses multi-layered neural networks. It is used for tasks such as image & speech recognition.

⑭ NLP = Natural Language Processing allows computers to understand & work human language.

Example are chatbots, translation apps & voice assistants.

Large

⑮ LLM = ^{Large} Language Model is an AI model trained on large amounts of text to understand & generate language. Examples include :- Deepsee, ChatGPT, Gemini

⑯ Generative AI Creates new Content Such as text, Images, audio or code based on learned patterns. Examples :- DALL-E; ChatGPT, Gemini;

- ①7 A training dataset is data used to teach an AI model. Good quality data is important because biased or incorrect data can cause inaccurate or unfair results.
- ①8 Computer vision allows computers to interpret images and video. Examples include facial recognition and self-driving car systems.
- ①9 An API allows different software systems to communicate with each other. It's like a waiter taking an order from a customer and communicating it to the kitchen. APIs allow systems to share data & services.
- ②0 An API key is a secret code used to identify & authenticate an application. It should be stored securely & never shared publicly.
- ②1 A REST API is a common way for systems to communicate over the web.
- GET :- Retrieve data
 - POST :- Create data
 - PUT :- Update data
 - Delete :- Remove data
- ②2 JSON is a lightweight format for storing & exchanging data. It is commonly used with APIs because it is easy for both humans & computers to read.
- Example :- `{ "name": "Name", "age": 30 }`

②③ Prompt Engineering is writing clear instructions to get better results from AI
Example

Prompt :- "Tell me about SQL"

Better :- Explain SQL to a beginner & give 5 Simple examples.

②④ A map function applies the same operation to every item in a list. Example :- $\text{map}(x \rightarrow x^2, [1, 2, 3])$
 $= [1, 4, 9]$

②⑤ Cloud computing provides computing resources over the internet. I.E :-
IaaS :- Virtual Servers (AWS)
PaaS :- Platform for building Apps (Google App Engine)
SaaS :- Software through net (Microsoft 365)

②⑥ Version Control tracks changes to files & codes over time. Git is widely used for this. A commit saves changes, while a branch allows separate work without changing the main code.

②⑦ Jupyter Notebook is an interactive environment used to write & run code, especially Python. It is popular because analysts can combine code, data, charts & explanations.

②⑧ Python is a popular programming language because it is relatively easy to learn & has many data & AI libraries.
Examples :- Pandas

Numpy

Matplotlib

Scikit-learn

(29) Data privacy means protecting people's information and controlling how it is collected and used.
GDPR is a European data protection law that applies to organisations handling personal data of people in EU/EEA, even in some cases when organisations is outside Europe.

(30) Data ~~privacy~~ Ethics
Means using data and AI responsibly and fairly.
Examples :- Using Someone's data without consent
Creating biased AI systems
Selling personal information without permission.

(31) Data Governance
Is a system of rules and responsibilities for managing data. It includes data quality, security, privacy, access and compliance.

(32) AI bias occurs when AI systems produce unfair results because of biased data or design. For example, a hiring system trained on biased historical data could unfairly favour one group over another.

(33) Data Careers
Role

Data Analyst :- Analyses data & creates reports

Data Scientist :- Builds models & finds complex patterns

Data Engineers :- Builds data pipelines & systems

AI Engineers :- Builds & deploys AI models

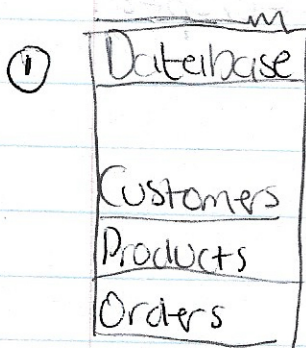
③④ Business Intelligence (BI)

Uses data, reports & dashboards to help businesses make decisions. Two popular tools * Power BI
* Tableau.

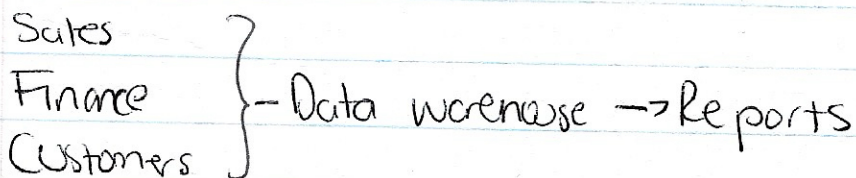
③⑤ I found generative AI the most interesting because it goes beyond text, it creates images & other content at a rapid speed

③⑥ Machine learning sounds quite difficult; the different types of models.

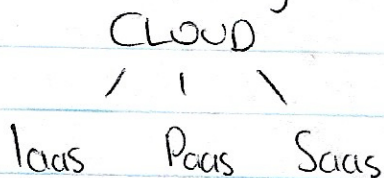
③⑦ I work with a lot of data and in a healthcare sector. I believe there is more for me to achieve and dive into with the knowledge I will get from here.



② Data Warehouse



③ Cloud Computing



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